

**MASTER OF COMPUTER
APPLICATIONS
(MCA-NEW)**

Term-End Examination

June, 2025

MCS-227 : CLOUD COMPUTING AND IoT

Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the
rest.*

1. (a) Define Cloud Computing. List and explain the *four* categories of Cloud Deployment models. 10

- (b) Define Virtual Machine (VM) sizing.
Explain any *one* way to do VM sizing. 10
- (c) What is 'Edge Computing' ? Explain its working. 10
- (d) Discuss the challenges in IoT Application Development. 10
2. (a) What is meant by 'Resource Sharing' in Cloud Computing ? Explain its implementation of single tenancy and multi-tenancy types of Resource Sharing in Cloud Computing. 10
- (b) List and explain any ***ten*** IoT technologies. 10

3. (a) Define a sensor with reference to an IoT device. Explain its characteristics. Explain any *two* IoT sensors along with their functions in the IoT devices. 10
- (b) What is Fog Computing ? Also, explain Desktop Level Virtualization. 10
4. (a) Write and explain fixed amount auto-scaling algorithm. 10
- (b) What is an Actuator ? Explain any *four* types of Actuators. 10
5. (a) What is a Cloud Computing Service Delivery Model ? Explain any *four* cloud computing service delivery models. 10

(b) What is meant by 'Load Balancing' ?

Explain Weighted Round Robin
algorithm with reference to load
balancing. 10

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December, 2024**

MCS-227 : CLOUD COMPUTING AND IOT

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Weightage : 70%

Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the rest.*

1. (a) What are the key characteristics that define cloud computing, and how to they differentiate cloud services from traditional computing model ? 10

- (b) How does virtualization operate at the machine or server level ? What are the key distinctions among full virtualization, para-virtualization and hardware-assisted virtualization ? 10
 - (c) What are the benefits of Security as a Service (SECaaS) ? How does SECaaS enhance overall cloud security ? 10
 - (d) Compare and contrast popular open-source IoT platforms. Highlight key features, scalability and customization capabilities to assist organizations in making informed decisions when selecting a platform for their IoT projects. 10
2. (a) What are the effective strategies to mitigate security issues in IoT environments ? Briefly discuss approaches such as encryption, secure device management and regular software updates

to ensure the resilience of IoT system
against cyber threats. 10

(b) What are the key features of a smart
grid ? How do these features contribute to
the overall efficiency and resilience of
energy distribution system ? 10

3. (a) Differentiate between Infrastructure as a
Service (IaaS), Platform as a Service
(PaaS) and Software as a Service (SaaS) in
cloud computing. 10

(b) What are the fundamental differences
between static and dynamic resource
provisioning ? How do these approaches
address the challenges of resource
allocation and scalability in cloud
computing environment ? 10

4. (a) How does auto-scaling contribute to optimizing resource utilization in the context of both vertical scaling and horizontal scaling ? 10
- (b) How does the concept of edge computing relate to the IoT and what advantages does it offer in terms of data processing and response time for IoT application ? 10
5. Write short notes on the following : 4×5=20
- (i) IoT security challenges
 - (ii) Horizontal *vs.* Vertical scaling
 - (iii) Fog *vs.* Edge computing
 - (iv) Technologies used for M2M Communication

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Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the rest.*

1. (a) Define Cloud Computing. List and explain the four categories of cloud deployment models. 10
- (b) Define resource sharing in cloud computing. Explain the implementation of single tenancy and multi-tenancy types of resource sharing in cloud computing. 10

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- (c) What is Internet of Things (IoT) ? What are the characteristics ? Also explain industrial IoT, infrastructure IoT and internet of military things (IoMT) categories of IoT. 10
- (d) Explain the differences between Fog computing and Edge computing. Draw the block diagram of 3-layer architecture of fog computing and explain all its layers. 10
2. (a) Differentiate between cluster, grid and cloud computing with respect to its characteristics, physical structure, hardware, resources, applications, networking and scalability features. 10
- (b) What is Auto scaling in cloud ? Write and explain fixed amount auto scaling algorithm. 10

3. (a) Define resource pooling in cloud environment. In this context, explain the following : 10

- (i) Server Pools
- (ii) Storage Pools
- (iii) Network Pools

- (b) With the help of a block diagram, explain the 4-levels in a cloud architecture. 10

4. (a) Define virtualization. Explain its underlying abstraction. Also mention the features provided by virtualization environment. 10

- (b) Explain the following communication protocols with reference to the IoT devices :

10

- (i) IPv6

(ii) MQTT

(iii) CoAP

(iv) XMPP

5. Write short notes on the following : $4 \times 5 = 20$

(a) Multihoming and its types

(b) Horizontal scaling in cloud environment

(c) Challenges in cloud computing

(d) Applications of edge computing

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Note : Question no. 1 is **compulsory** and carries 40 marks. Answer any **three** questions from the rest.

1. (a) What is resource provisioning in cloud computing ? Explain the static and dynamic approaches of resource provisioning. Mention their advantages and disadvantages. 10
- (b) Define Load Balancing. Explain the following algorithms with reference to load balancing : 10
- (i) Static algorithm approach
 - (ii) Weighted Round Robin

- (c) Discuss the following baseline technologies of IoT : 10
- (i) Security in IoT
 - (ii) IoT Analytics
 - (iii) IoT Processors
 - (iv) IoT Standards and Ecosystems
- (d) Define Edge Computing. Draw a block diagram of Cloud-Fog-Edge collaboration and explain all its layers. 10
2. (a) What is scalability in Cloud Computing ? Explain the following strategies of scaling : 10
- (i) Proactive Scaling
 - (ii) Reactive Scaling
- (b) Define VM (Virtual Machine) sizing. Discuss the two ways to do VM sizing. 5
- (c) Mention any five applications of Cloud Computing. 5
3. (a) Define a sensor with reference to an IoT device. Explain various characteristics of sensors. Also, mention and explain any four classifications of sensors. 10
- (b) Explain the following computing components used in laboratories of IoT/Cloud : 10
- (i) Arduino
 - (ii) Raspberry Pi

4. (a) Discuss the following Service Delivery Models of Cloud, with an example for each : 10

- (i) Platform as a Service (PaaS)
- (ii) Infrastructure as a Service (IaaS)

(b) “Cloud Computing offers a variety of deployment models, a network connection viewpoint will be used to examine Cloud deployment models and their accessible components.” With reference to this statement, discuss the following types of network connectivities : 10

- (i) Public Inter Cloud Networking
- (ii) Private Intra Cloud Networking

5. Write short notes on the following : 4×5=20

- (a) Applications of Fog Computing
 - (b) Desktop Level Virtualization
 - (c) Actuator and its any three types
 - (d) Zigbee, NFC and RFID connectivity technologies used in IoT applications
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Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the rest.*

1. (a) What is cloud computing service delivery models ? List and explain any *four* service delivery models. 10

(b) Explain the following scaling strategies : 10

(i) Proactive scaling

(ii) Reactive scaling

(iii) Combinational scaling

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Also, compare proactive and reactive scaling with reference to the suitability, working, cost and implementation parameters.

- (c) Explain all the *four* components, which support IoT system, with the help of a sample block diagram. 10
- (d) What is fog computing ? Explain the working of its along with any use case. Also, mention its advantages. 10
- 2. (a) Define edge computing. Also, briefly explain the working of edge computing. 10
- (b) What is Virtualization ? Explain its underlying abstraction. Also, mention its important characteristics. 10
- 3. (a) Compare and contrast cluster, grid and cloud computing with reference to characteristics, physical structure, hardware, resources, applications, networking and scalability parameters. 10

- (b) What is load balancing ? What is its functionality ? Explain the following load-balancers along with their features : 10
- (i) Network load balancer
 - (ii) Application load balancer
4. (a) Define a sensor with reference to an IoT device. Explain its characteristics. List and explain any *four* IoT sensors along with their functions in the IoT devices. 10
- (b) What is cloud security ? What are the threats to cloud security ? Explain various information security methods in cloud computing. 10
5. Write short notes on the following : 4×5=20
- (a) Public and private clouds
 - (b) Challenges in cloud computing
 - (c) Hierarchical structure of cloud
 - (d) Use case of edge computing

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Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the rest.*

1. (a) With reference to the Cloud Architecture, explain the following layers : 10
 - (i) Client access layer
 - (ii) Internet connectivity layer
 - (iii) Cloud service management layer
 - (iv) Layer of physical resources

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- (b) What is load-balancing ? Why is it imperative in cloud computing to balance the cloud-load ? Also, explain briefly hardware-based load balancer and virtual load balancer. 10
 - (c) List and explain any *ten* emerging IoT technologies. 10
 - (d) What is Edge Computing ? Explain its working with the help of a use-case. 10
2. (a) Explain the following IoT-connectivity technologies : 10
- (i) Zigbee
 - (ii) Z-wave
 - (iii) RFID
 - (iv) NFC
- (b) With reference to cloud-security, explain the following : 10
- (i) Identity management
 - (ii) Access control mechanism

3. (a) Explain any *five* applications of cloud computing. 5
- (b) Write a short note on utility computing. 5
- (c) Explain the evolution of cloud computing. 5
- (d) Describe briefly the Hybrid-Cloud Deployment Model. 5
4. (a) What is machine or server level virtualization ? Briefly explain the following types of server level virtualization : 10
- (i) Hypervisor
- (ii) Full-virtualization
- (b) Explain the following sensors used in IoT devices along with its features and applications : 10
- (i) Image sensors
- (ii) Chemical sensors
- (iii) Acceleration sensors
- (iv) Proximity sensors

5. Write short notes on the following : $4 \times 5 = 20$

- (a) Actuators and its types
- (b) MQTT, CoAP and XMPP IoT protocols
- (c) Challenges of IoT
- (d) Applications of fog computing

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Note : (i) *Question No. 1 is compulsory.*

(ii) *Answer any **three** questions from the rest.*

1. (a) Define Type-1 and Type-2 hypervisors in virtualization, highlighting their differences and use cases. How do they contribute to effective resource utilization in cloud environment ? 10
- (b) What are the main characteristics and advantages of public, private and hybrid

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cloud deployment models ? How does an organization decide which development model is most suitable for its specific business requirements ? 10

(c) What are the primary advantages that an organization can derive from adopting cloud computing ? Additionally, Could you offer real-world examples illustrating how business leverage cloud services to enhance efficiency, scalability and flexibility ? 10

(d) What do you mean by Edge-Computing ? How does it differ from cloud and Fog computing ? In what scenarios does edge computing Excel ? 10

2. (a) What are the key challenges in Fog-Computing ? Explain the commonly used strategies and technologies development to overcome security, scalability, and resource management, by ensuring the reliability of Fog computing environments. 10

- (b) What are the primary security challenges associated with IoT devices and networks ? What strategies and technologies are essential for addressing these challenges and ensuring a secure IoT environment ?
10
3. (a) What are the application and key challenges of IoT in smart transportation ? How does IoT contribute to improving efficiency; safety and sustainability in modern transport systems ?
10
- (b) What is scaling in cloud computing ? Describe Proactive and Reactive scaling strategies.
10
4. (a) Explain the architecture of Resource pooling. How are server, storage and network pools structured in cloud environments ? Discuss the significance of each pool in achieving efficient resource utilization.
10

- (b) What do you mean by Resource Sharing ?
Explain tenancy at different level of cloud services. 10

5. Write short notes on the following : $4 \times 5 = 20$

- (a) Scaling Strategies
- (b) Resource Provisioning
- (c) IoT security
- (d) Grid computing *vs.* Cloud computing